Spatial confounding and misaligned covariates in SDALGCP2

Restricted spatial regression, and covariates measured on a different support from the outcome

SDALGCP2 development notes

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Abstract

Two practical problems in areal disease mapping with spatially structured predictors. First, a spatially smooth covariate is collinear with the spatial random effect, which absorbs its signal and biases the regression coefficient (*spatial confounding*); we describe the restricted spatial regression implemented in `SDALGCP2`. Second, the covariate may be observed on a *different spatial support* from the outcome – other polygons, point monitors, or a coarser grid; we derive a change-of-support treatment that predicts the covariate to the candidate points feeding the intensity-scale offset and propagates the prediction uncertainty through a Berkson correction.

1 Background: the SDA-LGCP model

For aggregated counts Y_i over regions A_i, $i = 1, \ldots, N$, the SDA-LGCP sets

$$Y_i \mid \boldsymbol{\eta} \sim \text{Poisson}(m_i e^{\eta_i}), \qquad \boldsymbol{\eta} = \mathbf{D}\boldsymbol{\beta} + \mathbf{Z}, \qquad \mathbf{Z} \sim \mathcal{N}(\mathbf{0},\ \sigma^2 \mathbf{R}(\phi)), \tag{1}$$

where $\mathbf{D}$ ($N \times p$) is the region-level design, m_i an offset, and $\mathbf{R}(\phi)$ the aggregated spatial correlation built from candidate points inside each region.

2 Part I: spatial confounding

2.1 The problem

When a covariate column of $\mathbf{D}$ is spatially smooth, it lies close to the span of the leading modes of $\mathbf{R}(\phi)$. The random effect $\mathbf{Z}$ can then reproduce much of that column, so the data are equally well explained by "covariate effect" or by "spatial noise". The fitted $\widehat{\boldsymbol{\beta}}$ is pulled away from the value a non-spatial GLM would give, usually *attenuated* toward zero, with inflated variance. In `SDALGCP2` on a simulated gradient covariate (true $\beta = 0.5$) the standard fit returns $\widehat{\beta} \approx 0.28$ while a Poisson GLM returns ≈ 0.42 – the spatial term has eaten the signal.

2.2 Restricted spatial regression (RSR)

Following Reich, Hodges & Zadnik (2006) and Hughes & Haran (2013), constrain the random effect to the orthogonal complement of the fixed-effect design. Let $\mathbf{K}$ ($N \times r$, $r = N - p$) be an orthonormal basis of $\text{null}(\mathbf{D}^\top)$, so

$$\mathbf{K}^\top \mathbf{K} = \mathbf{I}_r, \qquad \mathbf{D}^\top \mathbf{K} = \mathbf{0}. \tag{2}$$

(`SDALGCP2` takes $\mathbf{K}$ from the trailing columns of a complete QR of $\mathbf{D}$.) Replace the unrestricted random effect by $\mathbf{K}\boldsymbol{\alpha}$:

$$\boxed{Y_i \mid \boldsymbol{\alpha} \sim \text{Poisson}(m_i e^{\eta_i}), \quad \boldsymbol{\eta} = \mathbf{D}\boldsymbol{\beta} + \mathbf{K}\boldsymbol{\alpha}, \quad \boldsymbol{\alpha} \sim \mathcal{N}\big(\mathbf{0},\ \sigma^2\,\mathbf{K}^\top \mathbf{R}(\phi)\mathbf{K}\big).} \tag{3}$$

Because $\mathbf{K}\boldsymbol{\alpha} \in \text{col}(\mathbf{D})^\perp$, the random effect *cannot* reproduce any column of $\mathbf{D}$; $\boldsymbol{\beta}$ is identified by the data, not by the spatial structure, and the spatial term only smooths the residual. RSR removes the confounding by construction; in the Gaussian case $\widehat{\boldsymbol{\beta}}$ equals the OLS estimate, and for the Poisson model it is close to the GLM estimate.

2.3 Estimation: Laplace-approximate marginal likelihood

Rather than Monte Carlo, (3) is fitted by integrating $\boldsymbol{\alpha}$ out at its mode. With $\boldsymbol{\Sigma}_{\boldsymbol{\alpha}}(\phi) = \mathbf{K}^\top\mathbf{R}(\phi)\mathbf{K}$ and $g(\boldsymbol{\alpha}) = \sum_i \{y_i\eta_i - m_i e^{\eta_i}\} - \frac{1}{2\sigma^2}\boldsymbol{\alpha}^\top {\boldsymbol{\Sigma}_{\boldsymbol{\alpha}}}^{-1}\boldsymbol{\alpha}$, let $\widehat{\boldsymbol{\alpha}} = \arg\max_{\boldsymbol{\alpha}} g(\boldsymbol{\alpha})$ (a few Newton steps; the negative Hessian is $\mathbf{H} = \mathbf{K}^\top \operatorname{diag}(m_i e^{\eta_i})\mathbf{K} + {\boldsymbol{\Sigma}_{\boldsymbol{\alpha}}}^{-1}/\sigma^2$). The Laplace-approximate marginal log-likelihood is

$$\ell(\boldsymbol{\beta}, \sigma^2, \phi) = \sum_i \{y_i\widehat{\eta}_i - m_i e^{\widehat{\eta}_i}\} - \tfrac{1}{2}\Big(r\log\sigma^2 + \log|\boldsymbol{\Sigma}_{\boldsymbol{\alpha}}| + \tfrac{\widehat{\boldsymbol{\alpha}}^\top {\boldsymbol{\Sigma}_{\boldsymbol{\alpha}}}^{-1}\widehat{\boldsymbol{\alpha}}}{\sigma^2}\Big) - \tfrac{1}{2}\log|\mathbf{H}| + \text{const}, \tag{4}$$

maximised over $(\boldsymbol{\beta}, \log\sigma^2)$ for each ϕ on a grid, with standard errors from the Hessian of (4). On the simulation above RSR returns $\widehat{\beta} \approx 0.44$ (SE 0.07) – de-confounded – while still estimating the spatial parameters. Use `sdalgcp(..., control = sdalgcp_control(confounding = "restricted"))`.

2.4 Spatio-temporal restricted regression

With the same regions observed at times $t = 1, \ldots, T$, stack the NT region–times and let $\mathbf{D}$ ($NT \times p$) be the space–time fixed-effect design and $\mathbf{K}$ ($NT \times (NT - p)$) an orthonormal basis of $\text{null}(\mathbf{D}^\top)$. The restricted space–time model keeps (3) but with the *separable* space–time covariance,

$$\boldsymbol{\eta} = \mathbf{D}\boldsymbol{\beta} + \mathbf{K}\boldsymbol{\alpha}, \qquad \boldsymbol{\alpha} \sim \mathcal{N}\Big(\mathbf{0},\ \sigma^2\,\mathbf{K}^\top\big(\mathbf{R}_t(\nu) \otimes \mathbf{R}_s(\phi)\big)\mathbf{K}\Big), \tag{5}$$

where $\mathbf{R}_s(\phi)$ is the aggregated spatial correlation and $\mathbf{R}_t(\nu)$ a temporal Matérn correlation. Because $\mathbf{K}\boldsymbol{\alpha} \in \text{col}(\mathbf{D})^\perp$ over all region–times, no spatial *or* temporal mode of the random effect can reproduce a column of $\mathbf{D}$, so $\boldsymbol{\beta}$ is de-confounded exactly as before. Estimation is the same Laplace-approximate marginal likelihood (4) with $\boldsymbol{\Sigma}_{\boldsymbol{\alpha}} = \mathbf{K}^\top(\mathbf{R}_t(\nu) \otimes \mathbf{R}_s(\phi))\mathbf{K}$, profiled over the spatial grid ϕ and maximised over the temporal range ν alongside $(\boldsymbol{\beta}, \sigma^2)$. The construction reduces *exactly* to the spatial RSR (3) at $T = 1$ (then $\mathbf{R}_t \equiv 1$ and ν drops out), a check used in the package tests. It forms the full $NT \times NT$ covariance, so it is $O\big((NT)^3\big)$ and best suited to modest NT. Use `SDALGCP2_ST(..., confounding = "restricted")`.

3 Part II: covariates on a different support

3.1 The problem

The intensity-scale treatment of a continuous covariate needs its value $z(x_{ik})$ at the candidate points x_{ik} inside each outcome region, entering the offset $b_i(\boldsymbol{\beta}) = \log\sum_k w_{ik}\exp\{z(x_{ik})^\top\boldsymbol{\beta}\}$ (companion note on raster covariates). A raster gives z everywhere, so this is immediate. But often the covariate is observed on a *different support*:

1. **point** locations $s_1, \dots, s_L$ (air-quality monitors, weather stations);
2. **other polygons** $B_1, \dots, B_M$ (census tracts, catchments) that do not nest with the outcome regions A_i;
3. a **coarser/finer grid** than the candidate points.

All three are *spatial misalignment* (change-of-support) problems (Gotway & Young, 2002; Gelfand, 2010): we must transfer z to the points x_{ik}.

3.2 Predict the covariate, then aggregate

Model the covariate as a latent spatial process $z(\cdot)$ and predict it to the candidate points from the observed data $\mathbf{z}^{\text{obs}}$.

Point support. Fit a geostatistical (Gaussian-process) model to $\{z(s_l)\}$ and krige to each x_{ik}, giving a predictive mean and variance

$$\mu_z(x_{ik}) = \mathbb{E}[z(x_{ik}) \mid \mathbf{z}^{\text{obs}}], \qquad v_z(x_{ik}) = \text{Var}[z(x_{ik}) \mid \mathbf{z}^{\text{obs}}]. \tag{6}$$

Areal support. If z is observed as averages over polygons B_j, $z_j^{\text{obs}} = |B_j|^{-1} \int_{B_j} z(u)\, du$, model the underlying surface $z(\cdot)$ and use *areal kriging*: the observations are linear (integral) functionals of z, so the joint Gaussian model gives the same predictive μ_z, v_z at the points x_{ik} – exactly the discrete-aggregation machinery `SDALGCP2` already uses, applied to the covariate.

Grid support. A special case of point/areal kriging; bilinear interpolation is the cheap plug-in.

3.3 Berkson correction: propagate the prediction uncertainty

Plugging the kriged mean $\mu_z(x_{ik})$ straight into $b_i(\boldsymbol{\beta})$ ignores that $z(x_{ik})$ is *predicted*, not observed – a classic errors-in-variables problem that attenuates $\boldsymbol{\beta}$. Because the kriging predictive distribution is Gaussian, $z(x_{ik}) \mid \mathbf{z}^{\text{obs}} \sim \mathcal{N}(\mu_z, v_z)$, we can integrate it out in closed form. Using $\mathbb{E}[e^{a^\top X}] = \exp(a^\top \mu + \frac{1}{2} a^\top V a)$ for $X \sim \mathcal{N}(\mu, V)$,

$$\mathbb{E}\big[\exp\{z(x_{ik})^\top \boldsymbol{\beta}\} \,\big|\, \mathbf{z}^{\text{obs}}\big] = \exp\Big\{\mu_z(x_{ik})^\top \boldsymbol{\beta} + \tfrac{1}{2}\, \boldsymbol{\beta}^\top V_z(x_{ik})\, \boldsymbol{\beta}\Big\}, \tag{7}$$

so the offset that correctly accounts for the misalignment uncertainty is

$$\boxed{b_i(\boldsymbol{\beta}) = \log \sum_k w_{ik} \exp\Big\{\mu_z(x_{ik})^\top \boldsymbol{\beta} + \tfrac{1}{2}\, \boldsymbol{\beta}^\top V_z(x_{ik})\, \boldsymbol{\beta}\Big\}.} \tag{8}$$

Equation (8) is the ordinary intensity-scale offset evaluated at the predicted covariate mean, *plus* a $\frac{1}{2}\boldsymbol{\beta}^\top V_z \boldsymbol{\beta}$ term that inflates the contribution of poorly-observed locations – the Berkson correction. It reduces to the raster case when $V_z \to 0$ (the covariate is known), and it is fitted by the same Gauss–Newton scheme as the raster model, with the effective covariate gradient $\partial b_i / \partial \boldsymbol{\beta}$ now including the $V_z \boldsymbol{\beta}$ term. Full propagation of the covariate model's own parameter uncertainty (a two-stage or joint model) is a natural further step.

Summary. Misaligned covariates are handled in two steps that reuse the existing machinery: (i) predict z to the candidate points by (areal) kriging, obtaining μ_z, v_z; (ii) feed them into the Berkson-corrected offset (8). The candidate-point grid that already discretises the outcome regions doubles as the common support on which outcome and covariate are aligned.

Spatio-temporal misalignment. When the outcome is observed over several times but the covariate support is time-invariant (e.g. fixed monitors), the covariate is kriged once to the candidate points and the Berkson offset (8) feeds the Kronecker-free space–time fit exactly as in the spatial case, inside the same Gauss–Newton loop (`SDALGCP2_ST(..., covariates = )`).

References. Reich, B.J., Hodges, J.S., Zadnik, V. (2006). Effects of residual smoothing on the posterior of the fixed effects in disease-mapping models. *Biometrics* 62, 1197–1206.
Hughes, J., Haran, M. (2013). Dimension reduction and alleviation of confounding for spatial generalized linear mixed models. *JRSS-B* 75, 139–159.
Gotway, C.A., Young, L.J. (2002). Combining incompatible spatial data. *JASA* 97, 632–648.